

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE - 1

COMPOSITION OF ATMOSPHERE

- **GASES**
- **WATER VAPOUR**

CLIMATOLOGY MODULES

Atmosphere	▪ Composition and structure; weather and climate;
Insolation	▪ Factors controlling insolation; heat budget of the earth — heating and cooling of atmosphere (conduction, convection, terrestrial radiation, advection);
Temperature	▪ Factors controlling temperature; distribution of temperature — horizontal and vertical; inversion of temperature;
Pressure	▪ Pressure belts; winds-planetary, seasonal and local; air masses and fronts; tropical and extra-tropical cyclones;
Precipitation	▪ Evaporation; condensation — dew, frost, fog, mist and cloud; rainfall — types and world distribution;
World Climate and Climatic Changes	▪ Different types of Climate

CLIMATE vs WEATHER




Weather Forecast
 21 May 2017

30°C
Chandigarh
 Chandigarh, India

Max: 43°C Min: 27 °C

CLIMATE

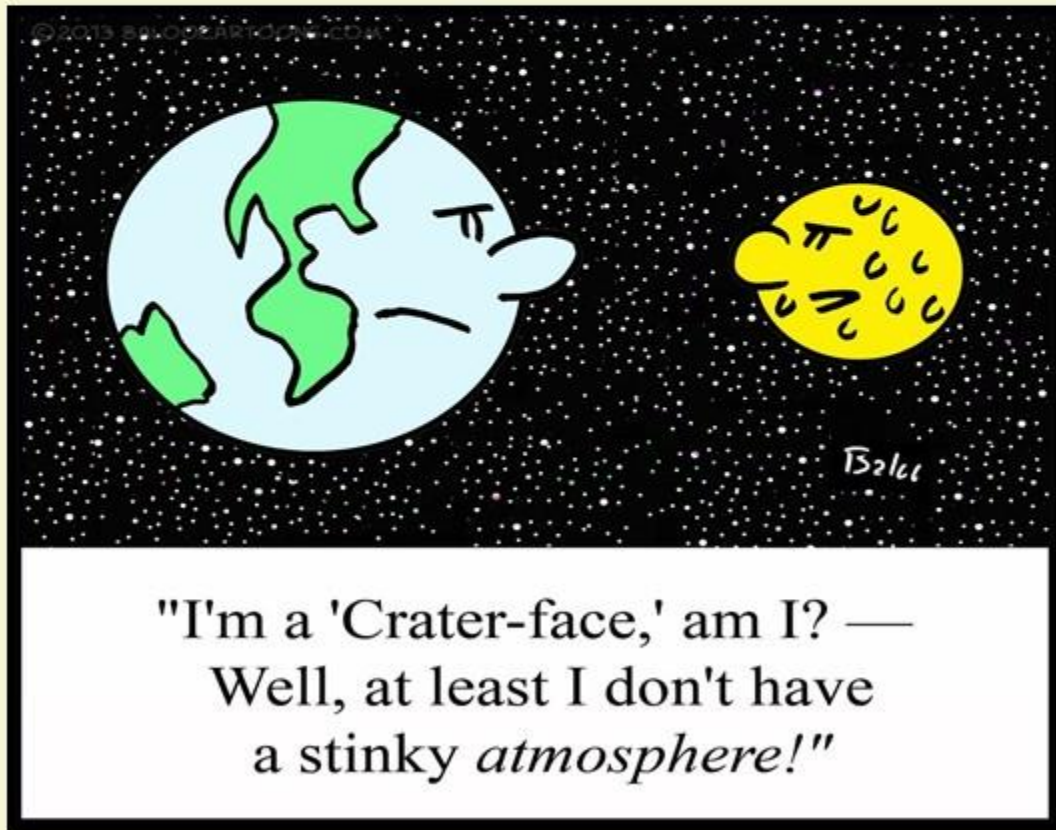
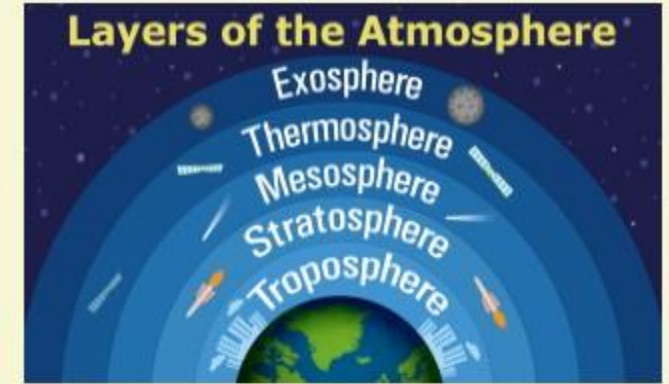
- It is defined as the average condition expected at a particular place in a **given timeframe**.
- Calculated by using the sum total of weather statistics over a time block of 30 years.
- “What you expect”
- **The study of climate in totality is known as CLIMATOLOGY.**

WEATHER

- It is defined as the condition of atmosphere **at a particular point of time**.
- Timely measurements of pressure of atmosphere, speed and direction of wind, rainfall, humidity, overcasts and other varying factors.
- “What you get”
- The study of weather, on the other hand, is known as **METEOROLGY.**

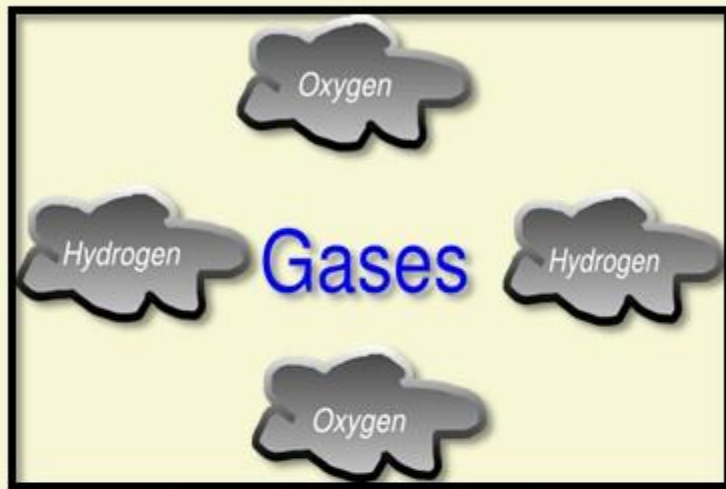


WHAT IS ATMOSPHERE?



- Atmosphere is a thick gaseous envelope that surrounds the Earth.
- Crucial for the biospheric ecosystem. Without this, earth would have become **barren like moon**.

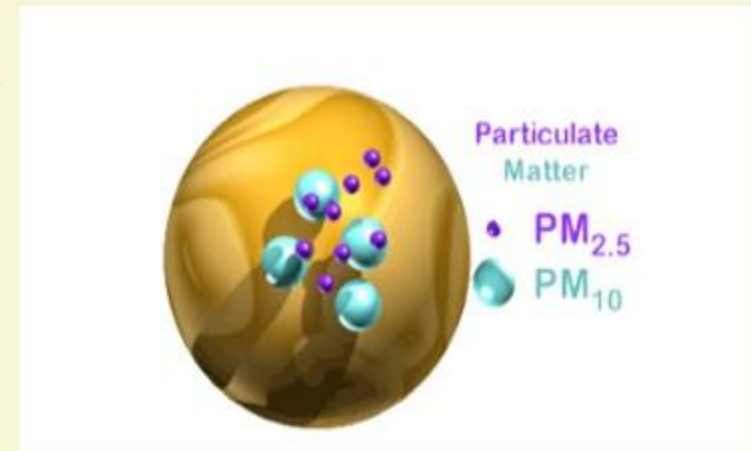
CONSTITUENTS OF ATMOSPHERE



1. GASES



2. WATER VAPOUR



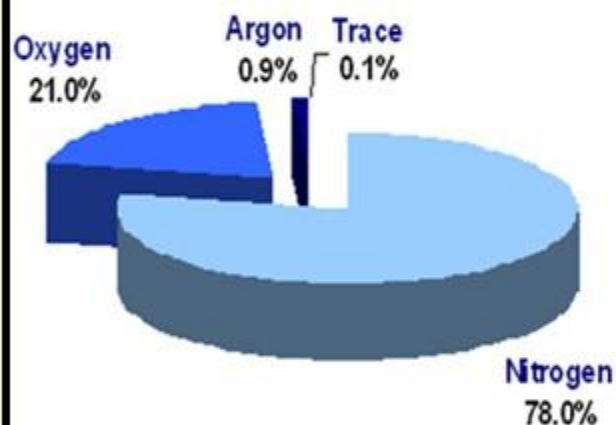
3. PARTICULATE MATTER



CONSTITUENTS OF ATMOSPHERE

1. GASES

Atmospheric Composition



Gases in the atmosphere

Gas	%
Oxygen	21
Carbon Dioxide	0.04
Nitrogen	78
Argon	0.9

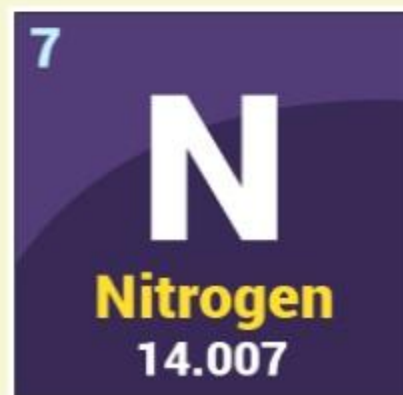
1 Oxygen is the most important gas for **living organisms**.

2 Nitrogen acts as **diluent** and it is chemically inactive.

3 Carbon dioxide is used by plants for **photosynthesis**.

What is the purpose of N_2 in the atmosphere?

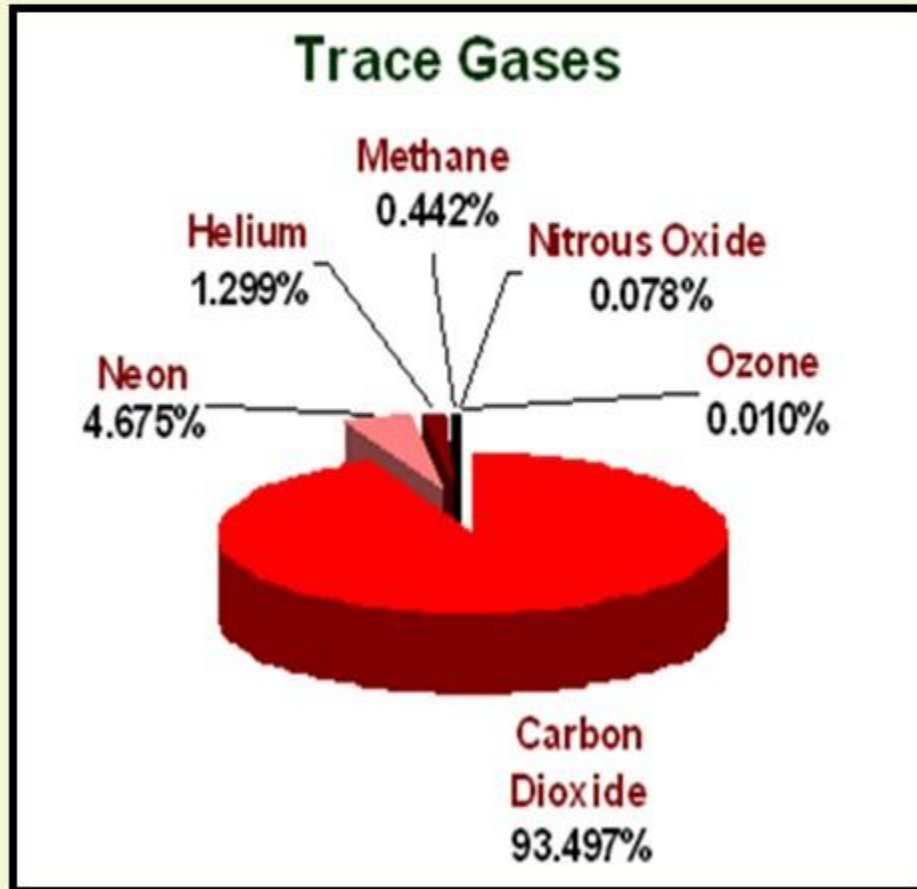
- To dilute oxygen as it is a **highly combustible gas**
- Human anatomy is designed in such a way that it cannot inhale 100% O_2 . The **presence of N_2** makes the air around us fit for breathing.



CONSTITUENTS OF ATMOSPHERE

1. GASES

TRACE GASES



1 Several atmospheric trace gases such as ozone, oxides of nitrogen and oxides of sulphur are anthropogenic.

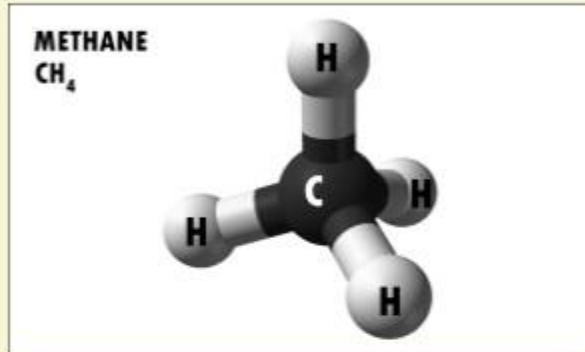
2 Others such as carbon dioxide and methane are important greenhouse gases and are produced anthropogenically and also by plants and animals.

CONSTITUENTS OF ATMOSPHERE

1. GASES

TRACE GASES

1. METHANE



Top sources of methane are:

1. Agriculture (E.g.: flooded rice field)
2. Digestion and gas emission of animals like cows
3. Wetlands
4. Fossil fuel production
5. Biomass burning
6. Urban landfills

CONSTITUENTS OF ATMOSPHERE

1.GASES

TRACE GASES

1.METHANE

FOSSIL FUEL PRODUCTION



BIOMASS BURNING



FARM ANIMALS



AGRICULTURE



WETLANDS



URBAN LANDFILLS



CONSTITUENTS OF ATMOSPHERE

1.GASES

TRACE GASES

2.OZONE

Vienna Convention for the Protection of the Ozone Layer



Kigali makes history with HFC freeze

India among 197 nations to phase out the family of greenhouse gases within various timelines

JACOB KOSHY

NEW DELHI: Countries came to an agreement in Kigali on Saturday to phase out a family of potent greenhouse gases that results

NOT A COOL REFRIGERANT



India follows up Paris treaty ratification with promise to phase out Hydrofluorocarbons

Q: What is HFC-23?

It is a potent greenhouse gas that results

Q) What's happening in Kigali?

198 countries agreed to phase out gases



Q) What is India's plan?

It will "freeze" HFC use by 2028 on a 2024-2026 "baseline", which means it will

CONSTITUENTS OF ATMOSPHERE

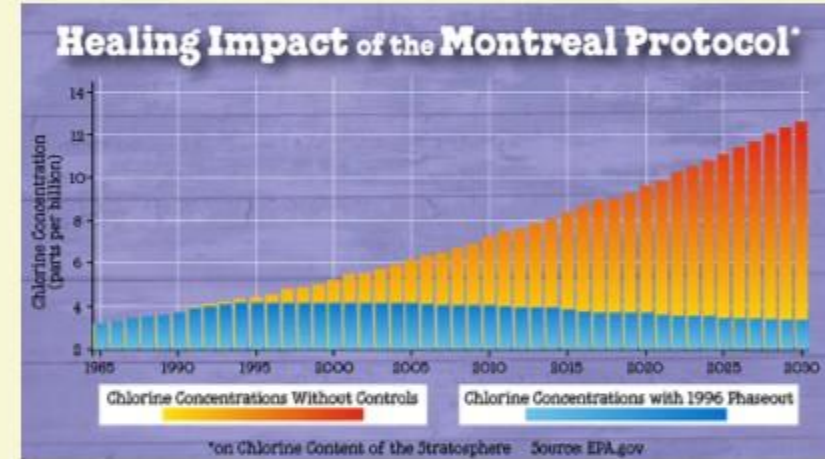
1.GASES

TRACE GASES

2.OZONE

WHY IS THIS IMPORTANT FOR UPSC?

- The Vienna Convention for the Protection of the Ozone Layer entered into force in 1988. Hence this year is its 30th anniversary.



BEWARE!!

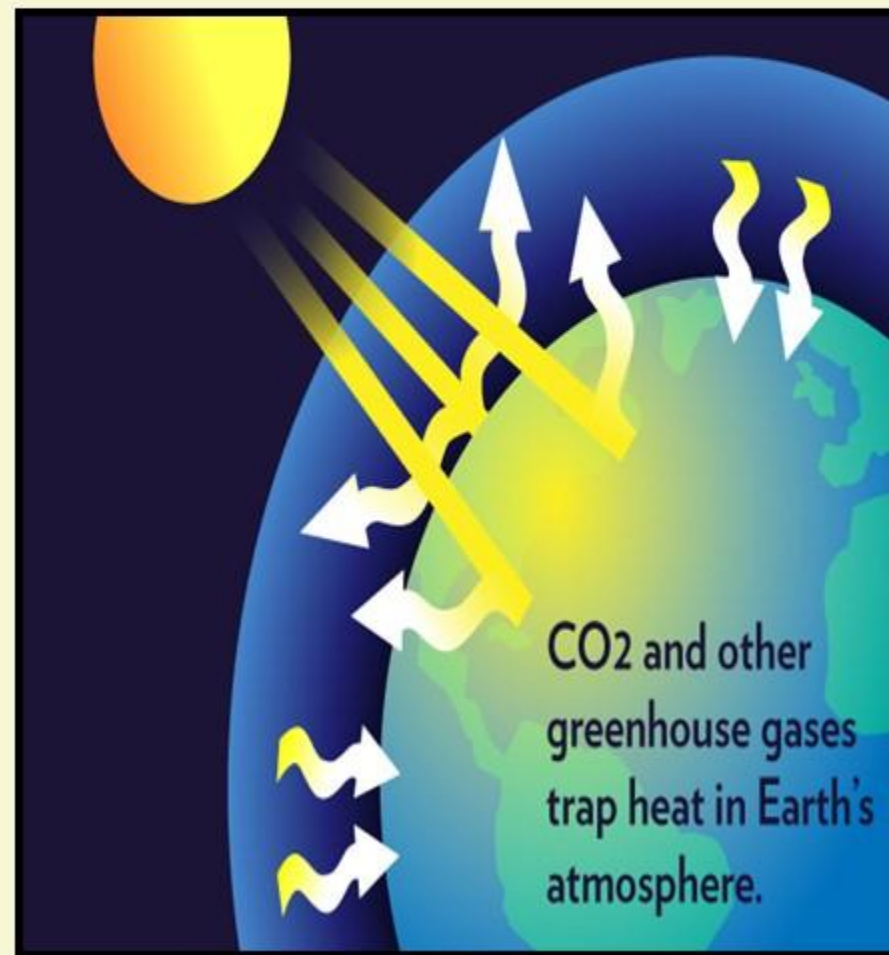
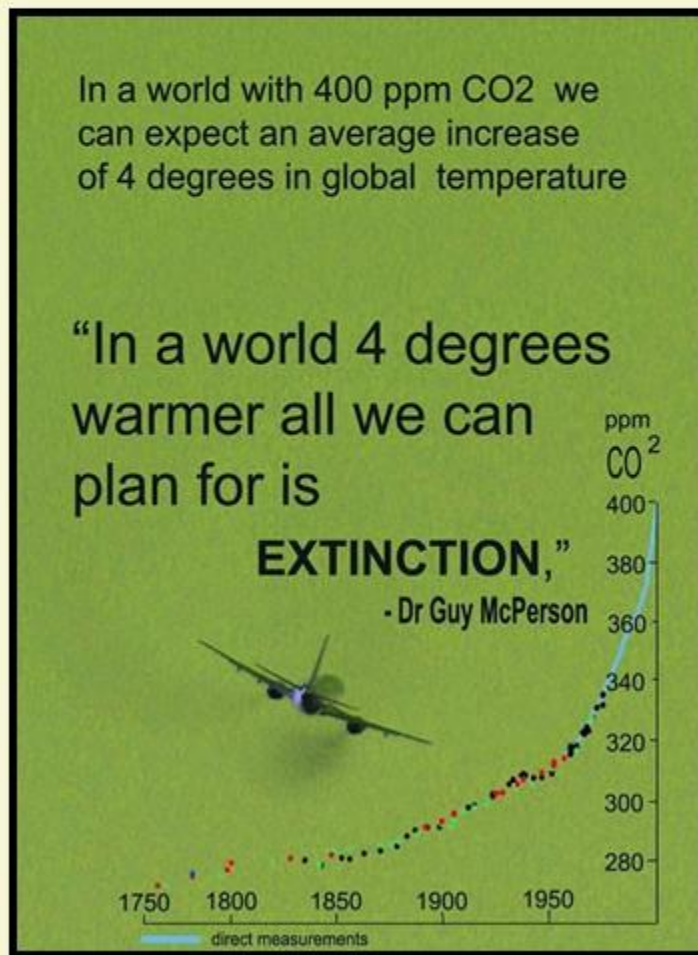
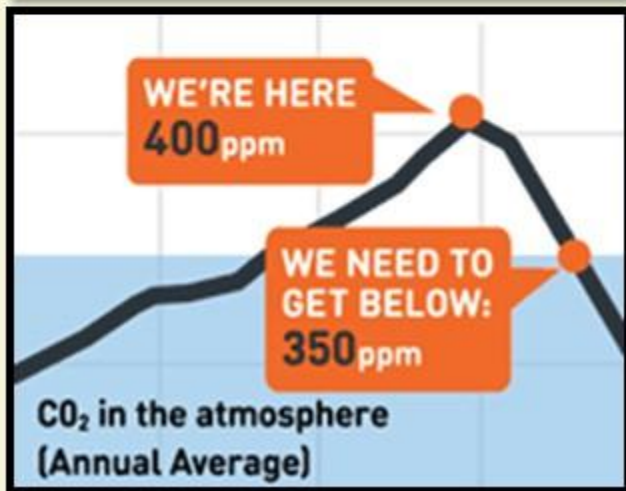
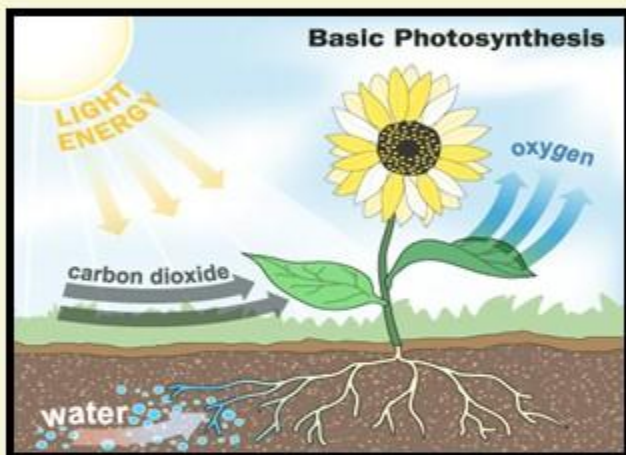
However, Vienna Convention **does not** include legally binding reduction goals for the use of CFCs, the main chemical agents causing ozone depletion. These are **laid out** in the accompanying Montreal Protocol.

CONSTITUENTS OF ATMOSPHERE

1. GASES

TRACE GASES

3. CARBON DIOXIDE



CONSTITUENTS OF ATMOSPHERE

1. GASES

TRACE GASES

4. SO_x

SOURCES OF SO_x

- 1 Coal-fired Power plants and fossil fuel combustion
- 2 Volcanic eruptions
- 3 Manufacture of sulphuric acid
- 4 conversion of wood pulp to paper
- 5 smelting of mineral ores (aluminum, copper, zinc, lead, and iron)

IMPACTS OF SO_x

- 1 Acid rain killing lifeforms
- 2 Particulate soot
- 3 Long-term exposure can cause breathing problems like asthmatics, whereas short-term exposure can lead to chest tightness and coughing and wheezing.



CONSTITUENTS OF ATMOSPHERE

1.GASES

TRACE GASES

5.NO_x

SOURCES OF NO_x

- 1 Vehicular exhaust and fossil fuel combustion
- 2 Produced naturally by bacteria and agricultural fertilisation
- 3 Lightning
- 4 Nitric acid manufacture
- 5 Welding processes
- 6 Use of explosives

IMPACTS OF NO_x

- 1 Acid rain killing lifeforms
- 2 Photochemical smog



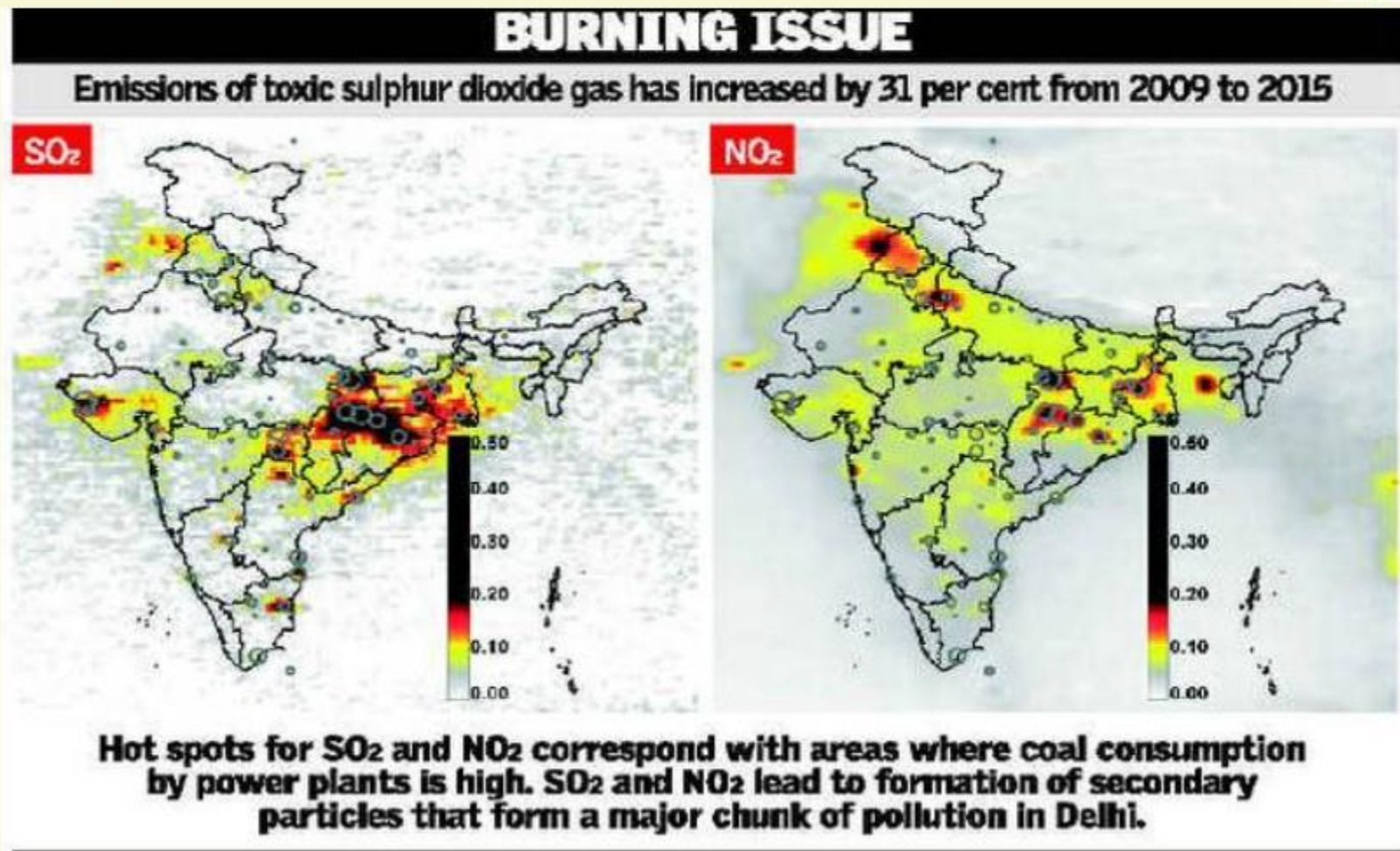
TRACE GASES

SO_x & NO_x

The report titled “**Out of Sight: How coal burning advances India’s air pollution crisis**” by Greenpeace India used satellite imagery from 2009 to 2015 to find that the below shown areas had thermal power plants and they were reason behind the growth in emissions of SO_2 and NO_2 .



GREENPEACE



Large black spots=High Coal Consumption=Large Thermal Power Plants



CONSTITUENTS OF ATMOSPHERE

1.GASES

TRACE GASES

Q14. Consider the following:

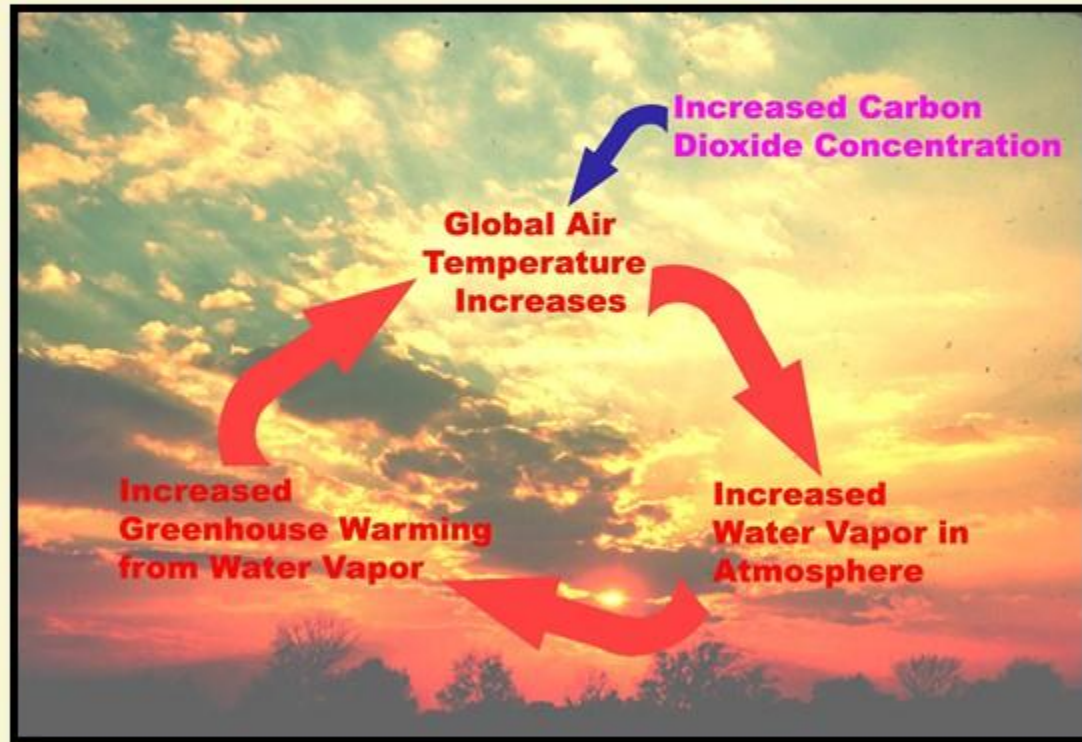
1. Carbon dioxide
2. Oxides of Nitrogen
3. Oxides of Sulphur

Which of the above is/are the emission/emissions from coal combustion at thermal power plants ? (**UPSC Prelims 2011**)

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3**

CONSTITUENTS OF ATMOSPHERE

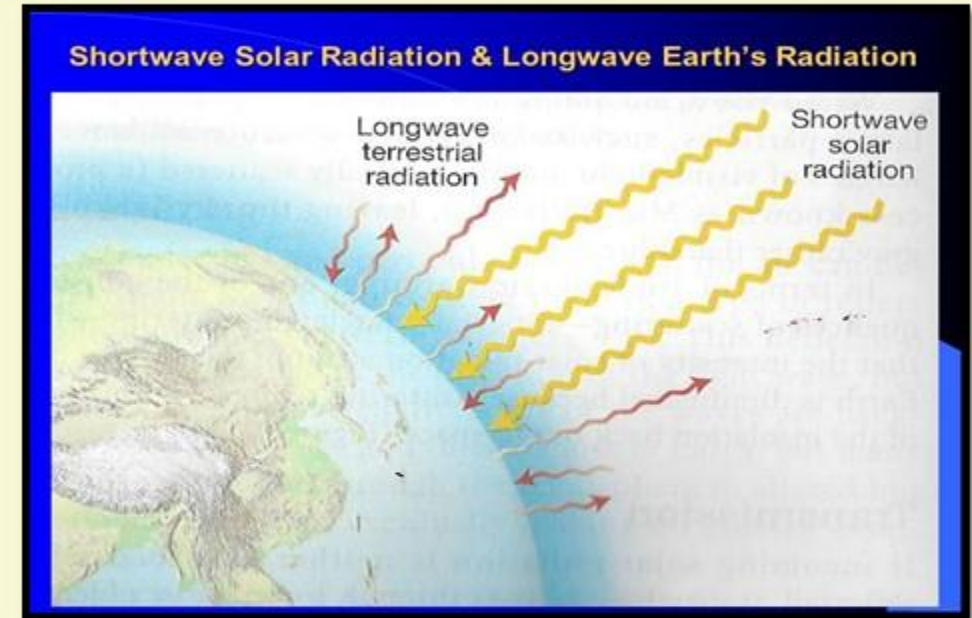
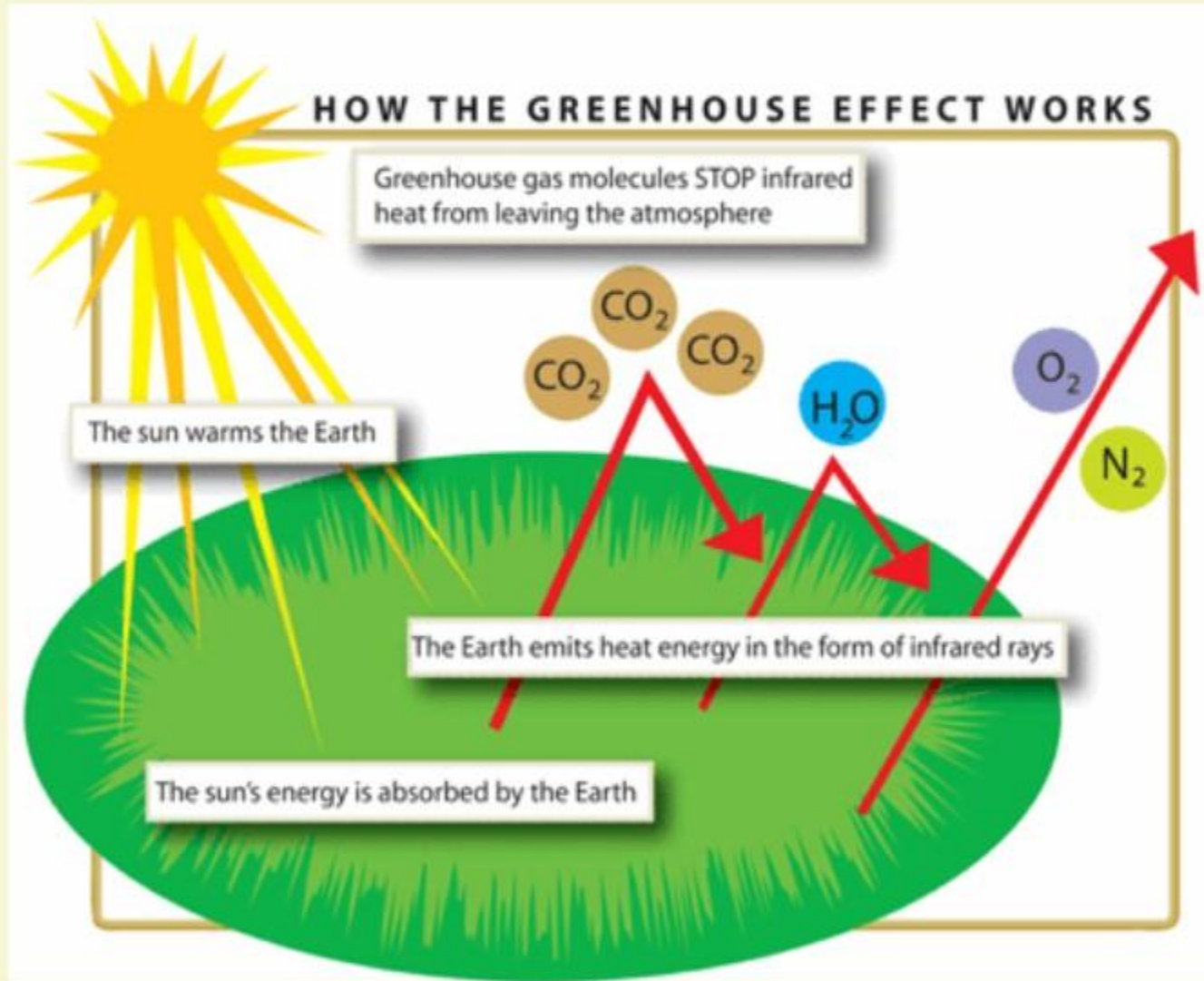
2. WATER VAPOUR



- More than 90% of the total atmospheric vapour is found upto the height of 5km.
- Vapour depends on temperature.
- More the evaporation (as in tropical areas) more will be the vapour content.
- Responsible for trapping the outgoing long-wave radiation.

CONSTITUENTS OF ATMOSPHERE

2. WATER VAPOUR



WATER VAPOUR AND CO_2 ARE BOTH GREENHOUSE GASES!!

Transparent to the incoming shortwave radiation and traps the outgoing long-wave radiation. This phenomenon is **greenhouse effect**.

CONSTITUENTS OF ATMOSPHERE

1.GASES

POSSIBLE MAINS QUESTION

Topic: GS Paper 1 : Urbanization- their problems and remedies

GS Paper 3 : Environmental pollution and degradation

It is an interesting paradox that though US and China are emitting greenhouse gases still Delhi is rated as one of the most polluted city on the Earth. Discuss.

(10 Marks) (200 words)

HINTS

1

Biggest greenhouse gas emitters

US and China

CO₂ and CH₄

Hazardous for Climate

Pollution emitters

Cities like Delhi

NO_x gases and Particulate Matter

Hazardous for Health

2

Developed vs Developing country phenomena

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 - 2 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9